

Grid transformation for the world's largest energy projects

- Best-in-class energy density and round-trip efficiency
- Industry-leading power electronics and thermal system performance
- Rapid and cost-effective deployment with factory-assembled and pre-tested solution

Scaled and rigorously tested product safety and reliability

- Comprehensive in-house reliability testing by the leading experts in the industry
- Engineered for safety and performance at every level
- Continuous improvement based on large-scale operational experience

Designed with flexibility and configurability in mind

- Modular architecture that allows for a range of configurations across multiple applications
- Industry experts available to identify site-specific needs
- Integrated solution that allows for battery augmentation over time

**POWER AND ENERGY**

Megapack duration is configurable. Standard configurations are 2-Hour and 4-Hour durations. Nominal energy is specified at 25°C (77°F).

	AC Power per Megapack	Energy per Megapack
2-Hour	1927 kW	3854 kWh
4-Hour	979 kW	3916 kWh

ELECTRICAL

Nominal AC Voltage	480 V AC 3-phase	
Nominal Frequency	50 or 60 Hz	
Inverter Power per Megapack¹	2-Hour Max:	2400 kVA
	4-Hour Max:	1320 kVA
Round-Trip Efficiency²	2-Hour:	92.0%
	4-Hour:	93.7%

¹ Scalable from 400 kVA minimum in increments of 50 kVA

² Full-depth cycle including all power conversion and thermal system losses, at 25°C (77°F)

WARRANTY

Coverage	All-inclusive, equipment and energy retention
Term	15 years standard, extendable to 20 years

PART NUMBER

1848844-XX-Y Where X is a number between 0-9 and Y is a letter

MECHANICAL AND MOUNTING

Ingress Ratings	IP66/NEMA 3R (Main Enclosure) IP20 (Thermal System)		
Enclosure Dimensions	Width:	8800 mm	(346 ½ in)
	Depth:	1650 mm	(65 in)
	Height:	2785 mm	(110 in)
	+/- 13 mm (½ in)		
Maximum Weight	38,100 kg (84,000 lb)		
Operating Ambient Temperature	-30°C to 50°C (-22°F to 122°F)		

REGULATORY

System is compliant to grid codes and safety standards of all major markets.

System	NRTL listed to UL 1973, UL 9540, UL 9540A, UL 1741 SB, IEC 62619, IEEE 1547
Cells	NRTL listed to UL 1642

CONTROLS AND COMMUNICATIONS

Protocols	Modbus TCP / DNP3 / REST API	
Core Control Modes	Direct Real Power	Ramp Rate Control
	Direct Reactive Power	Site Control
	Frequency Support	Power Factor Control
	Virtual Inertia	Voltage Control

MONITORING

Powerhub	Free-to-use cloud monitoring portal
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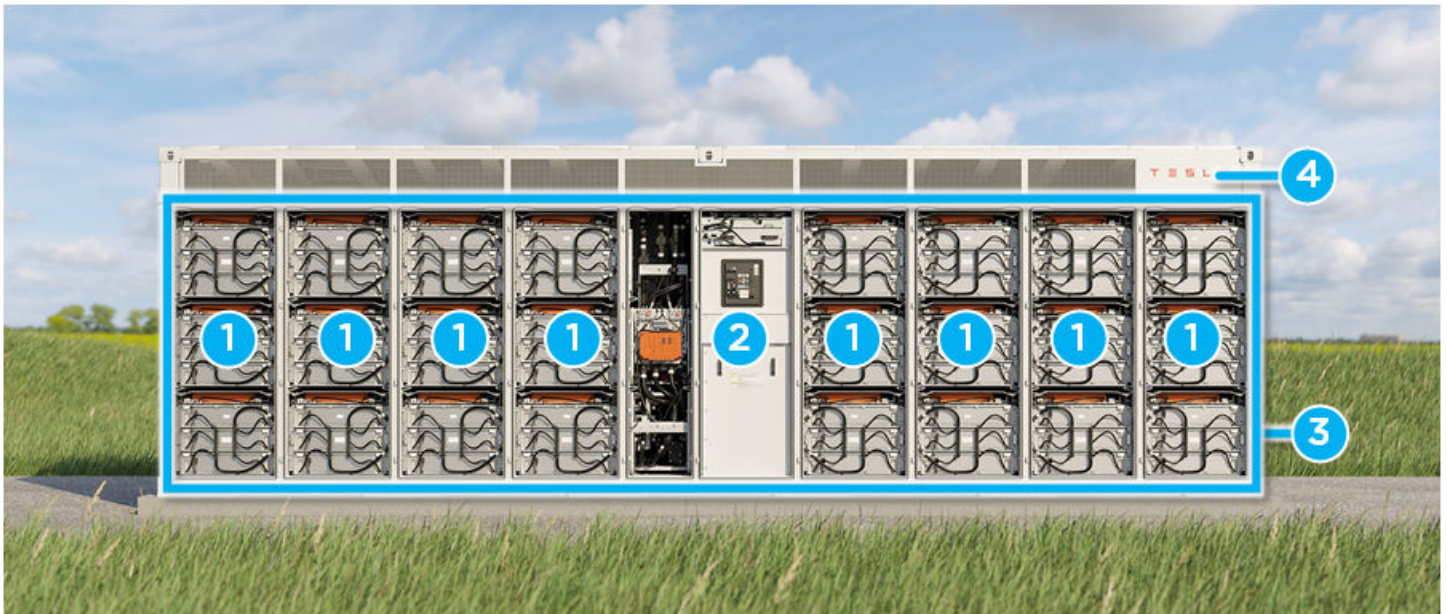
MEGAPACK 2 XL SAFETY OVERVIEW

ENHANCED SAFETY ARCHITECTURE

Tesla's commitment to safety informs every Megapack design decision and has guided 15+ years of experience in battery module design and manufacturing for both vehicle and energy storage applications. Megapack 2 XL (Megapack) is designed with features that make the product safe throughout the entire product lifecycle — during transit, installation, commissioning, operation, maintenance, and decommissioning.

Tesla's approach to safety involves comprehensive design and testing at every level of Megapack. Vertical integration across design, manufacturing, and testing ensures that safety features of the cell, battery module, inverter, thermal system, and overall system-level components are closely linked and not decoupled.

In addition, Tesla is continually improving Megapack safety features and capabilities based on data from operational experience.



1. Battery modules with active and passive fuses – externally serviceable
2. Touch-safe Customer Interface Bay
3. Non-walk-in IP66 enclosure and deflagration mitigation
4. Thermal roof with overpressure vents

INDUSTRY-LEADING COMPLIANCE AND THIRD-PARTY VALIDATION

Tesla is constantly pushing the boundaries and raising the bar on product safety. This commitment to safety not only ensures that Tesla's products are compliant to the industry's most stringent global standards, but also sets a benchmark for the industry to follow regarding energy storage safety. Megapack has met and exceeded many industry safety standards and has demonstrated through extensive third-party testing that it is one of the safest energy storage systems on the market.



MEGAPACK 2 XL SAFETY OVERVIEW

Megapack 2 XL is listed to the following standards by OSHA-recognized Nationally Recognized Testing Laboratories:

- UL 1642 (cell-level certification)
- UL 1973 and IEC 62619 (battery module-level certification)
- UL 9540, IEC 62933-5-2, IEC 62109-1 (system-level certification)
- UL 1741, CSA C22.2 #107.1 (power electronics)
- UL 1998 and IEC 60730 Annex H (functional safety of software)
- IEC 61000-6-2, and EN 55011 (EMC)
- UN 38.3 (transportation, self-certified)
- IEEE 693 (seismic safety)
- UL 9540A (large-scale fire testing): Tested at the cell, module, and unit level
- And many more, including compliance to major market grid codes

Megapack 2 XL, like Megapack, is designed to comply with major installation codes for energy storage systems, including NFPA 855, IFC 2018 and 2021, and NEC 2020.

Megapack 2 XL has been reviewed and validated by an Independent Engineer, both at the product level and for the results of large-scale fire testing.

ENHANCED APPROACH TO FIRE SAFETY

To date, Tesla has deployed more than 10 GWh of stationary energy storage products globally with a strong safety track record.

Through vertical integration, Tesla has designed Megapack with fire safety built directly into the product at every level. This makes the product safer and reduces overall project costs by eliminating the need for fire suppression systems.

At the cell level, Tesla's latest generation of Megapacks leverages the lithium iron phosphate (LFP) chemistry and a new industry-leading cell design. Testing has demonstrated a strong ability to resist thermal runaway, and has shown controlled venting in worst-case events, without explosive bursts or fire.

All Tesla products also undergo rigorous testing at the module level. While standards such as UL 1973 and IEC 62619 ensure propagation resistance to single-cell thermal runaway, testing has shown that Megapack battery modules are resistant to multiple co-located cells sent into runaway at the same time. This greatly mitigates the risk of a thermal event.

At the system level, Megapack is designed with a combination of dedicated runaway gas igniters and overpressure vents built into the roof that passively mitigate the risk of deflagration hazards in case of unlikely accumulation of flammable gases due to arc flash events or thermal runaways.

In the unlikely event of a fire, rigorous full-scale fire testing has shown that Megapack performs in a safe and controlled manner, consuming itself slowly and without explosive bursts, projectiles, or unexpected hazards. The vents are designed to direct all gases, smoke, and flame out of the top of the Megapack, minimizing risk to nearby response personnel and exposures.

In the event of a fire at a Megapack site, the fire service will be able to manage the event with standard fire service response equipment. Tesla's *Lithium-Ion Battery Emergency Response Guide* provides more details on that subject. The cells used in Tesla products do not contain solid metallic lithium and thus do not react with water. When required by local code, Tesla recommends fire detection at the site level with the use of third-party thermal imaging cameras that can detect fires on site.



MEGAPACK 2 XL SAFETY OVERVIEW

24/7 GLOBAL SUPPORT

Megapack is supported by Tesla's Network Operations Center, designed to support the global fleet of energy storage products. The 24/7 operations center offers remote monitoring, diagnostics, and troubleshooting capabilities, without the need of having a Tesla technician on site. Customers and first responders also benefit from immediate hotline support from trained technicians in case of emergencies.



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Rev 1.0 — September 13, 2022



MEGAPACK SAFETY APPLICATION NOTE

This application note is intended to complement the *Megapack Safety Datasheet*. This document provides additional technical detail regarding Megapack's core safety features and thermal runaway hazards and should be used for technical discussions with customers and AHJs.

An Integrated Approach to Safety

When it comes to safety and product hazards, Tesla's philosophy has always been to design products that are safe to customers, maintenance personnel, first responders and neighboring exposures, regardless of the operating status of the product. Megapack is safe by design and includes numerous levels of safety and hazard mitigation features that make it one of the safest utility-scale energy storage systems (ESS) in the world. Extensive testing and iterative design improvements have demonstrated this approach, validated by a growing fleet of more than 6 GWh of energy storage products deployed globally. In addition, Tesla is continually improving Megapack safety features and capabilities based on data from operational experience.

Safety at Every Level

Cell Safety

- Tesla uses the highest quality lithium-ion cells, which are designed specifically for the application of the products they go into. Each cell in Tesla products is individually inspected and tested prior to module assembly. Flawed cells are discarded and used to drive further cell manufacturing improvements. All cells in Tesla products are listed to UL 1642.
- Each Megapack can contain upwards of 200,000 cells. Each cell is individually fused through wire bond and is equipped with its own internal circuit interrupt device (CID). The CID ensures that any overcurrent or unsafe pressure buildup within a cell is isolated to that cell. Both features ensure the cell is physically and irreversibly disconnected from the rest of the module and does not impact the surrounding cells or modules for the remainder of the product's life.

Battery Module Safety

- Each Megapack battery module includes multiple active and passive safety features designed to isolate electrical issues from the rest of Megapack, and conversely to protect the battery module from external issues. Unlike racked battery systems, the battery modules are pre-installed in a factory setting, removing the need to connect live high voltage DC elements on site.
- Battery modules are listed to UL 1973 and IEC 62619 and are tested to ensure a single cell in thermal runaway does not propagate to neighboring cells. In fact, Tesla Megapack battery modules have demonstrated effectiveness at preventing thermal runaway propagation from multiple co-located cells set off into runaway at the same time.



- The battery modules each include a dedicated DC-DC converter that electrically and galvanically isolates the module from Megapack's common DC bus. Each Megapack can hold up to 17 battery modules, which are connected in parallel via a common DC bus. This parallel battery module architecture provides optimized performance and redundant safety control, reducing the risk of cascading failures. Combined with the DC-DC converter architecture in each battery module, this parallel design greatly reduces the risk of events such as electrical fault propagation, arc flash, or cascading thermal runaway. Battery modules are sealed to prevent touch access to the battery cells, power electronics, and HV terminals.
- A Tesla-proprietary smart *pyrotechnic disconnect* installed at the midpoint of each battery module adds an additional layer of safety to each module. The disconnect is actively controlled by the battery management system (BMS) housed within each battery module, which continuously searches for and autonomously detects conditions that fall outside of nominal current and voltage operating limits. If an issue is detected, within milliseconds the BMS automatically triggers a pyrotechnic discharge inside the disconnect that immediately opens the connection and removes the source of the issue. This is identical technology to that used in Tesla vehicles. The disconnect is mounted at the front of the battery module so it can easily be replaced in the field without having to remove the entire battery module.

System-Level Safety

Numerous dedicated safety measures ensure that Megapack provides the highest level of safety:

- The *Customer Interface Bay* is a touch-safe bay located at the center of the enclosure. It is designed to be the only interface required for installation, operation, and maintenance. Megapack is pre-assembled and pre-tested at the factory to ensure that minimal installation and commissioning scope is required on site, thereby minimizing risk of hazards to personnel interfacing with the equipment.
- An internal safety feature called an *enable circuit* connects to all Megapack doors except the Customer Interface Bay door. If a door is opened while the product is operating or in standby mode with the DC bus powered, the enable circuit will immediately open to stop all operation and power down the DC bus. The Customer Interface Bay contains a panel between the user and the bus bars that is also part of the enable circuit.
- Megapack's weatherproof steel enclosure is rated to IP66 (NEMA 3R) and provides robust protection against extreme environmental, chemical, and physical exposures. It cannot be entered by personnel, further limiting the possible interaction between maintenance personnel and internal components.
- Megapack is listed to UL 9540 and UL 1998 / IEC 60730, which are system-level safety certifications that ensure all subsystems are designed to operate safely together under normal and abnormal conditions. Megapack's design is fully vertically integrated and all components (all hardware ranging from cell to enclosure, and all software and firmware) are designed by in-house Tesla teams, which further ensures full control over all aspects of the product.

Autonomous Safeguards and Notification

Megapack is engineered to react to fault conditions in an autonomous manner, with safeguards built into the firmware. These fault conditions include over-temperature, loss of communication, over-voltage, and isolation. All Tesla products are monitored for correct operation by Tesla's Network Operations Center (NOC), which provides 24h/7d monitoring and phone/email technical engineering support across the global fleet of deployed projects.



- For inverter faults, isolation faults, and internal loss of communication that prevent the safe operation of the system, Tesla's monitoring system will automatically alert the Tesla NOC to initiate a corrective action, which can either be performed remotely or in person during a field service visit.
- If a propagating thermal runaway occurs, over-temperature faults isolate the concerned Megapack by first disconnecting the affected battery module, and then opening the AC contactors. All faults are immediately seen by the Tesla NOC for immediate action.

Thermal Event Safety

Tesla operates its own rigorous fire and abuse testing facility to test and validate every product it brings to market. Tesla performs extensive fire testing at all component levels during the product design cycle to ensure that hazards are mitigated and managed safely. For Megapack, although catastrophic damage due to fire is unlikely, it is essential—both for Tesla and its customers—to understand how the product behaves in a worst-case condition that falls outside the product's control.

Through testing Megapacks and with data from operational experience, Tesla has demonstrated that a propagating thermal runaway event from internal causes is unlikely due to physical internal separation elements and the battery module's design, which requires multiple co-located cells to go into runaway at the same time. Events caused by rare and unlikely internal factors or unforeseen external factors (for example, a large impact) that damage many cells at once could create a propagating thermal runaway event that may spread throughout the entire enclosure. In such a case, it is likely that all battery modules will safely consume themselves, as Megapack is not designed to prevent propagation across battery module bays.

Deflagration Control

While deflagration/explosion protection is not required for outdoor cabinet-based designs in the model Codes and Standards, Tesla firmly believes mitigation of the chance of a deflagration within the enclosure is critical to overall safety. The design of Megapack includes two proprietary engineered core features that mitigate the risk of uncontrolled deflagration: over-pressure vents and a sparker system. The proprietary engineered deflagration prevention system developed by Tesla relies upon early ignition of the gases produced by thermal runaway to prevent a dangerous building of gases within the enclosure.

Pressure-sensitive vents in the roof of the sealed enclosure allow gases and products of combustion to safely exhaust through the roof in the event of extreme hazards (see location in Figure 1, below). The vents are designed to mitigate the impact of over-pressure by directing all gases, smoke, and flame out of the top of Megapack and ensure the front doors remain shut for the safety of nearby exposures and personnel. The vents are passive and are not actuated or controlled in any way. Their rubber seals are designed to release over-pressure events (including arc flash events), or melt out during thermal runaway events. The number and total area of vents is sized off of NFPA 68-13 requirements, and designed to be a factor of safety >3 on the enclosure strength, including the front doors. Co-located with the vents are one-way "umbrella" valves that help exhaust and disperse runaway gases in order to help minimize gas concentrations inside the enclosure.

The vents are not actuated active deflagration vents and are thus not designed or listed to corresponding standards (such as NFPA 68/69). Tesla developed the proprietary vents and sparker system because the application of NFPA 68 or NFPA 69 is not appropriate for cabinets without large volumes of open space.



In combination with the roof vents, Tesla has designed a proprietary *sparker system* that combusts flammable off-gases during a runaway event before those gases reach the enclosure's lower flammability limit (LFL). Hot gases and smoke are the first sign of thermal runaway, and the sparkers can ignite these gases well before they are able to build up in dangerous quantities and become an explosion hazard. There are eight sparkers in total, installed at the top of each battery module bay directly under the over-pressure vents in the roof (see location in Figure 1, below). Testing has shown that the highest measured concentration of hot rising gases will always occur in these locations. By co-locating the sparkers with the roof vents, localized combustion events ensure the roof vents nearest the sparker open. The effectiveness of the sparkers and roof vents has been successfully demonstrated through extensive testing, as well as during the UL 9540A Unit level test. This system also effectively mitigates the risk of any large deflagration, explosions, projectiles, or flying debris.

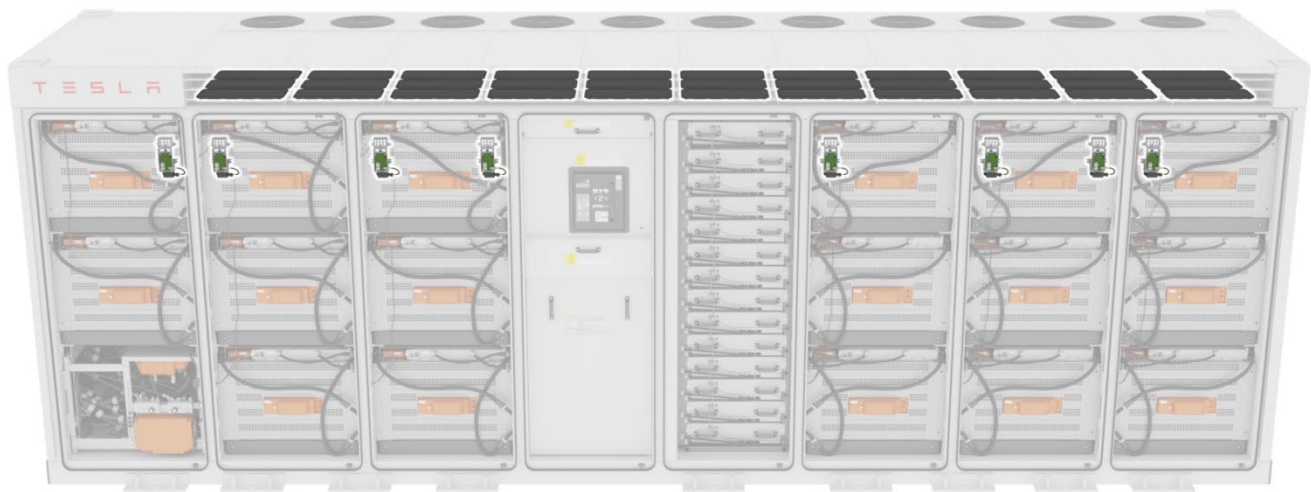


Figure 1: Roof vents and sparker locations inside Megapack

While the design of a sparker system is new and unique in the energy storage market, Tesla's goal was to use first-principles engineering design to solve the inherent risk of accumulated gases inside a sealed enclosure in an effective and consistent manner, knowing that a detection-only solution would not solve the problem.

Below are additional details on the sparker system's features and capabilities:

- The sparker system uses double platinum automotive spark plugs. The sparker assembly is designed to operate continuously (every 10 seconds) for 20+ years with minimal maintenance.
- The sparker system is always on and is powered by internal battery power, and thus will remain operational even during loss of grid power or if an external shutdown is triggered. It is connected to Megapack's low voltage distribution board in the Customer Interface Bay so it cannot be impacted from the first runaway event.
- The number and location of sparkers is designed in a redundant manner: should one sparker fail, another one is sufficiently close to effectively combust a localized gas concentration.
- The sparkers cannot combust the gases of a single cell failure. It requires the off-gassing of many co-located cells to reach sufficient localized LFL levels, thus the system is intended only to work during catastrophic irreversible thermal runaways.
- The sparkers do not interact with the outside environment, as they are located inside the sealed IP66 (NEMA 3R) enclosure. Should smoke or gases from an external or nearby event surround Megapack, the sparkers will be unable to combust those.



Thermal Insulation

To prevent propagation to neighboring Megapacks at 6 in (15 cm) spacing, Megapack includes several features that minimize the radiation of heat flux onto neighboring surfaces, as well as heat flux from external sources:

- 1 in (2.5 cm)-thick thermal insulation layer built into the sides and rear surfaces of the enclosure, covered in steel.
- Roof panels covered in high reflectivity paint and insulated internally with a layer of thermal insulation.
- Battery bay doors and ceiling internally layered with thermal insulation.

This thermal insulation protection has been validated through large-scale fire testing, including the UL 9540A Unit level test. During testing, cell temperatures in the neighboring Megapack's battery modules never exceeded the product's nominal operating temperature range, even with the thermal management system disabled. Also, during testing, the thermal runaway did not propagate to the neighboring units.

Fire Detection and Suppression

A product that is able to effectively control thermal event hazards greatly mitigates concerns over harm to people or property. For this reason, Tesla always designs products that burn in a safe, predictable manner and do not rely on complex and unproven systems to mitigate the hazards. The 'let it burn' approach also ensures that the system is not dependent on detection systems to mitigate major hazards and does not require any form of suppression.

Megapack does not have internal hardware dedicated to detecting thermal runaway (such as a smoke or off-gas detectors). Tesla does have the ability through software to detect a thermal runaway event and notify the Network Operations Center. External detection systems, while not required by model Codes and Standards for outdoor installations, are also a suitable option for customers who desire additional detection capability at the site level. Tesla has validated a pole-mounted third-party thermal imaging camera that can successfully detect early signs of thermal runaway well before flames are visible. Please contact Tesla for more details on this solution.

Megapack does not include any form of suppression system (water-based or non-water-based). While water is considered the most effective form of suppression for lithium-ion-based energy storage systems, Tesla's goal is to offer products that do not depend on water to manage the hazards of thermal events.. As such, Megapacks can be installed in remote locations without access to water (or where the fire service is not in proximity to the site) without worrying about the risk of expansive damage to the site should a catastrophic event occur to one unit.

Frequently Asked Questions

What happens if one cell fails and goes into thermal runaway?

Cell failures are rare, but at massive deployment scale they can occur. For this reason, the battery modules are tested to resist single cell thermal runaway propagation (UL 1973 and IEC 62619 requirement). If one cell goes into runaway inside Megapack, it will consume itself without propagating to neighboring cells. It will not impact the operation of the battery module, as the loss of one cell within the module brick is too small to cause imbalance in the system. The module can keep operating and the consumed cell will remain in its place without the need to be replaced. Additionally, a single cell runaway is insufficient to create enough gas concentration to be combusted by the sparkler system. The



gas will dissipate through the umbrella valves without impacting the rest of Megapack. Given the low impact of a lost cell, no alarms or alerts will be triggered by the loss of a single cell.

Can the thermal management system be utilized to mitigate thermal runaway?

The thermal management system in Megapack provides enormous cooling capacity to the cells and electronics. For the purposes of evaluating the worst case scenario, the thermal management system is always disabled during fire testing (for both the unit under test as well as neighboring units). This allows Tesla to understand how the system behaves in the event of thermal management system failures. However, should a thermal runaway occur in a real-world setting, Megapack will automatically react to the rise in cell temperature and enable the thermal management system to cool the cells as if it were operating normally. The thermal system will keep operating until loss of communications or the fire melts away controls and hardware components. Thermal management systems in neighboring Megapacks will continue to operate nominally and further cool the cells to keep them within safe temperature to avoid long-term damage.

What are the hazards of fires started from non-battery sources?

Fires from non-battery sources are much smaller in magnitude and release less embedded energy than battery fires, which are deep-seated chemical fires that generate a great quantity of heat. All power electronics in Megapack are separated from the battery modules by steel surfaces (either on top of the battery module, or in the Customer Interface Bay or inverter bay). If an electrical fire were to occur, it would remain contained in its dedicated enclosure and not generate sufficient heat to propagate to the cells. As demonstrated in Tesla's 2016 NFPA Research Foundation fire testing of Powerpack, setting cells into thermal runaway via external flame required over 40 minutes of direct contact from a 400kW flame against the steel enclosure (equivalent to about 50 backyard gas BBQs).

Would a coolant leak create the possibility for a deflagration within the enclosure?

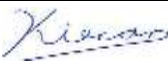
If all 540L of coolant leaked within the enclosure, there would still be not enough H₂ buildup to reach LFL at the enclosure level. In the event of a leak, coolant will evacuate the enclosure through the two bottom drains in the enclosure floor.

What improvements has Tesla made to Megapack in response to the incident at the Victoria Big Battery site in July 2021?

Tesla engaged in a rigorous investigation and root-cause analysis and implemented improvements into the product as a direct result of this event. Mitigating actions included a combination of firmware enhancements to safety protection features, improved monitoring alerting, and construction and commissioning process improvements. Through these actions, Tesla has reduced the likelihood of this rare event recurring in the future.

TEST REPORT	
ANSI/CAN/UL 9540A:2019	
Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems	
Report Number.....	NN22GLCI.001
Date of issue	04/26/2022
Total number of pages	28
Name of Testing Laboratory preparing the Report	TÜV Rheinland of North America, Inc. 1279 Quarry Lane, Suite A, Pleasanton, CA 94566
Applicant's name	Tesla, Inc.
Address	3500 Deer Creek Road, Palo Alto, CA 94304
Test specification:	
Standard	ANSI/CAN/UL 9540A:2019
Test procedure.....	Report
Non-standard test method	N/A
Test Report Form No.....	N/A
Test Report Form(s) Originator....	N/A
Master TRF	Dated 2019-01-17
General disclaimer:	
The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the Issuing Testing Laboratory. The authenticity of this Test Report and its contents can be verified by contacting the CB, responsible for this Test Report.	
Other / Scope:	
1. Evaluation and Compliance report to UL 9540A: 2019 – Unit level Round 2.	

Report No. NN22GLCI.001

Test item description	Battery Energy Storage System	
Trade Mark	Tesla	
Manufacturer	Tesla, Inc. (new # 1210368) 3500 Deer Creek Rd, Palo Alto, CA 94304	
Model/Type reference	1748844-XX-Y XX – can be any number from 00 to 99. XX – represents style codes used for different variants of the same part, having no impact on the safety and functionality of the entire product. Y – Can be any upper case letter from A to Z. Y – represents pedigree and is used for tracking changes to parts that have already been released to suppliers or production, having no impact on the safety and functionality of the entire product.	
Ratings	480Vac, 997.5 kVA (4hr), 1890 kVA (2 hr), 50/60 Hz (3 phase 3 wire) Nominal Battery capacity 4hr: 3100.8 kWh, 2 hr: 2890.8 kWh Nominal Battery power 4hr: 775.2 kW, 2 hr: 1445.4 kW	
Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):		
☒	Testing Laboratory:	TÜV Rheinland of North America, Inc. 1279 Quarry Lane, Suite A, Pleasanton, CA 94566
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
☐	Testing procedure: CTF Stage 1/TMP:	
Testing location/ address		
Tested by (name, function, signature)		
Approved by (name, function, signature) ..		
☒	Tesla, Inc.	Tesla, Inc.
Testing location/ address		Tesla Battery Test Facility Fernley, Nevada
Tested by (name + signature)		Kiran Krishnan Kutty 
Witnessed by (name, function, signature) .		Himanshu Vaidya 
Approved by (name, function, signature) ..		Howard Liu 
☐	Testing procedure: CTF Stage 3/SMT:	
☐	Testing procedure: CTF Stage 4:	
Testing location/ address		
Tested by (name, function, signature)		
Witnessed by (name, function, signature) .		
Approved by (name, function, signature) ..		
Supervised by (name, function, signature) :		

Report No. NN22GLCI.001

List of Attachments (including a total number of pages in each attachment):

1. Test package with testing equipment list
2. Photo documentation

Summary of testing:

Tests performed (name of test and test clause):

UL 9540A clause 9 – Unit Level

Testing location:

Tesla, Inc.
 Tesla Battery Test Facility
 Fernley, Nevada

Summary of compliance with National Differences (List of countries addressed): N/A

☐ The product fulfils the requirements of _____ (insert standard number and edition and delete the text in parenthesis, leave it blank or delete the whole sentence, if not applicable)

Report No. NN22GLCI.001

Possible test case verdicts: - test case does not apply to the test object.....: N/A - test object does meet the requirement.....: P (Pass) - test object does not meet the requirement.....: F (Fail)
Testing: Date of receipt of test item February 22, 2022 Date (s) of performance of tests March 09, 2022
General remarks: "(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a ☐ comma / ☒ point is used as the decimal separator.
Name and address of factory (ies) :
Copy of marking plate: Use – “Only for use with Tesla Products” “The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCB’s that own these marks”
General product information and other remarks: Megapack is bi-directional, supporting charge and discharge. It converts power for storage in rechargeable lithium-ion battery packs (battery modules) and is designed in a modular fashion in order to support a range of AC power. Each Megapack contains up to 19 battery modules with inverters, a thermal bay and associated thermal roof components, an AC circuit breaker, and a set of customer interface terminals and internal controls circuit boards. An external auxiliary power supply is not required for Megapack; Megapack pulls auxiliary power for the control power and thermal management from the local AC. Depending on the system configuration (2-hour or 4-hour), each Megapack can be configured with different quantities of battery modules which, together with the site's grid voltage, determine Megapack's nominal power rating.

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
CONSTRUCTION			--
5	General		--
5.1	Cell		
5.1.1	The cells associated with the BESS that were tested shall be documented in the test report	CATL Lithium Iron Phosphate Rated Capacity (Ah): 157.2Ah Nominal Voltage: 3.22V Nominal mass: 2991 ± 60 g Dimension: 50.75±0.5mm*166±0.5mm*169.3±0.5mm	
5.1.2	The cell documentation included in the test report shall indicate if the cells associated with the BESS comply with UL 1973	Battery module and cell is certified to UL 1973	P
5.1.3	Refer to 7.6.1 for further details	See 7.6.1	N/A
5.2	Module		
5.2.1	The modules associated with the BESS that were tested shall be documented in the test report	Battery module (3 modules in series, 336S1P)	P
5.2.2	The module documentation included in the test report shall indicate if the modules associated with the BESS comply with UL 1973	Battery module is compliant with UL 1973	P
5.2.3	Refer to 8.3 for further details	See 8.3	N/A
5.3	Battery energy storage system unit		P
5.3.1	The BESS unit documentation included in the test report shall indicate the units that comply with UL 9540	UL 9540 compliant and certified	P
5.3.2	For BESS units for which UL 9540 compliance cannot be determined,	See above	N/A
5.3.3	If applicable, the details of any fire detection and suppression systems that are an integral part of the BESS shall be noted in the test report	No fire detection and suppression systems used	N/A
5.3.4	Refer to 9.7, 10.4 and 10.7 for further details	See 9.7	N/A
5.4	Flow Batteries		N/A
5.4.1	For flow batteries, the report will cover the chemistry, as well as the electrical rating in capacity and nominal voltage of the cell stack	Not flow Batteries	N/A
5.4.2	The flow battery documentation included in the test report shall indicate if the flow battery system complies with UL 1973		N/A
5.4.3	See 7.6.2 for further details		N/A
PERFORMANCE			--
6	General		N/A

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
6.1	The tests in this standard are extreme abuse conditions conducted on electrochemical energy storage devices that can result in fires		N/A
6.2	At the conclusion of testing, samples shall be discharged in accordance with the manufacturer's specifications		N/A
9	Unit Level		
9.1	Sample and test configuration		P
9.1.1	The unit level test shall be conducted with BESS units installed as described in the manufacturer's instructions and this section. Test configurations include the following:		P
	a) Indoor floor mounted non-residential use BESS; b) Indoor floor mounted residential use BESS; c) Outdoor ground mounted non-residential use BESS; d) Outdoor ground mounted residential use BESS; e) Indoor wall mounted non-residential use BESS; f) Indoor wall mounted residential use BESS; g) Outdoor wall mounted non-residential use BESS; h) Outdoor wall mounted residential use BESS; and i) Rooftop and open garage non-residential use BESS installations.	Outdoor ground mounted non-residential use BESS	P
9.1.2	The unit level test requires one initiating BESS unit in which an internal fire condition in accordance with the module level test is initiated and target adjacent BESS units representative of an installation	One initiating BESS and three target adjacent BESS	P
	Exception: Testing can be conducted outdoors for outdoor only installations if there are the following controls and environmental conditions in place:	Testing can be conducted outdoors for outdoor only installations See Figure 1	P
	a) Wind screens are utilized with a maximum wind speed maintained at ≤ 12 mph; b) The temperature range is within 10°C to 40°C (50°F to 104°F); c) The humidity is $< 90\%$ RH; d) There is sufficient light to observe the testing; e) There is no precipitation during the testing; f) There is control of vegetation and combustibles in the test area to prevent any impact on the testing and to prevent inadvertent fire spread from the test area; and g) There are protection mechanisms in place to prevent inadvertent access by unauthorized persons in the test area and to prevent exposure of persons to any hazards as a result of testing.	This was an outdoor installation test. The ambient temperature was varied between 10°C and 15°C and humidity less than 90% RH, and wind was under 10 mph.	P
9.1.3	Depending upon the configuration and design of the BESS (e.g. the BESS is composed of multiple separate parts within separate enclosures), this testing to determine fire characterization can be done at the battery system level	Testing performed at BESS level	N/A
9.1.4	The initiating BESS unit shall contain components representative of a BESS unit in a complete installation.	Complete unit in the testing	P

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
9.1.5	Target BESS units shall include the outer cabinet (if part of the design), racking, module enclosures, and components		P
9.1.6	The initiating BESS unit shall be at the maximum operating state of charge (MOSOC),	100% SOC	P
9.1.7	If a BESS unit includes an integral fire suppression system, there is an option of providing this with the DUT	No integral fire suppression system	N/A
9.1.8	Electronics and software controls such as the battery management system (BMS) in the BESS are not relied upon for this testing.		P
9.2	Test method – Indoor floor mounted BESS units	Outdoor ground mounted units. Used the test method described in the Section 9.2 except conflicted with Section 9.3.	-
9.2.1	Samples and test configurations are in accordance with 9.1.	Testing conducted outdoor	N/A
9.2.2	Any access door(s) or panels on the initiating BESS unit and adjacent target BESS units shall be closed,	Doors closed	P
9.2.3	The initiating BESS unit shall be positioned adjacent to two instrumented wall sections	Wall construction per 9.3.3	N/A
9.2.4	Instrumented wall sections shall extend not less than 0.49 m (1.6 ft) horizontally beyond the exterior of the target BESS units.	Wall construction per 9.3.3	N/A
9.2.5	Instrumented wall sections shall be at least 0.61-m (2-ft) taller than the BESS unit height	Wall construction per 9.3.3	N/A
9.2.6	The surface of the instrumented wall sections shall be covered with 16-mm (5/8-in) gypsum wall board and painted flat black	Wall construction per 9.3.3	N/A
9.2.7	The initiating BESS unit shall be centered underneath an appropriately sized smoke collection hood of an oxygen consumption calorimeter	Testing conducted outdoor	N/A
9.2.8	The light transmission in the calorimeter's exhaust duct shall be measured using a white light source and photo detector for the duration of the test	Testing conducted outdoor	N/A
9.2.9	The chemical and convective heat release rates shall be measured for the duration of the test, using the methodologies specified in 8.2.11 and 9.2.12, respectively	Testing conducted outdoor	N/A
9.2.10	With reference to 9.2.9, the heat release rate measurement system shall be calibrated	Testing conducted outdoor	N/A
9.2.11	With reference to 9.2.9, the convective heat release rate shall be measured using thermopile	Testing conducted outdoor	N/A
9.2.12	With reference to 9.2.9, the convective heat release rate shall be calculated using the following equation: $HRR_c = V_e A \frac{353.22}{T_e} \int_{T_o}^T C_p dT$	Testing conducted outdoor	N/A
9.2.13	The physical spacing between BESS units (both initiating and target) and adjacent walls shall be representative of the intended installation	Wall construction per 9.3.3	N/A
9.2.14	Separation distances shall be specified by the manufacturer for distance between:		P

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
	a) The BESS units and the instrumented wall sections; and b) Adjacent BESS units.	a) 5 ft from Initiator to the left b) 6 inches from ISO knuckle of Initiating unit to Target unit. 4 inches from surface of initiating unit to target unit surface.	P
9.2.15	Wall surface temperature measurements shall be collected for BESS intended for installation in locations with combustible construction.	Wall construction per 9.3.3	N/A
9.2.16	Wall surface temperatures shall be measured in vertical array(s) at 152-mm (6-in) intervals for the full height of the instrumented wall sections using No. 24-gauge or smaller,	Wall construction per 9.3.3	N/A
9.2.17	Thermocouples shall be secured to gypsum surfaces by the use of staples placed over the insulated portion of the wires	Wall construction per 9.3.3	N/A
9.2.18	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each instrumented wall:	Wall construction per 9.3.3	N/A
	a) Both are collinear with the vertical thermocouple array;		N/A
	b) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module; and		N/A
	c) One is positioned at the elevation estimated to receive the greatest heat flux during potential propagation of thermal runaway within the initiating BESS unit.		N/A
9.2.19	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each adjacent target BESS unit that faces the initiating BESS unit:		P
	a) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module within the initiating BESS; and		P
	b) One is positioned at the elevation estimated to receive the greatest surface heat flux due to the thermal runaway of the initiating BESS.		P
9.2.20	For non-residential use BESS, heat flux shall be measured with the sensing element of at least one water-cooled Schmidt-Boelter gauge	Testing conducted outdoor	N/A
9.2.21	No. 24-gauge or smaller, Type-K exposed junction thermocouples shall be installed to measure the temperature of the surface	No. 24-gauge, Type-K used	P
9.2.22	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth	Non-residential	N/A

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
9.2.23	An internal fire condition in accordance with the module level test shall be created within a single module in the initiating BESS unit:	See Figure 2 Megapack can consist up to 19 Battery Module assemblies. Each module assembly contains 3 battery Modules each which is a total of 112 cell modules. Two sections of heaters for 2 cells was setup to force thermal runaway.	P
	a) The position of the module shall be selected to present the greatest thermal exposure		P
	b) The setup (i.e. type, quantity and positioning) of equipment for initiating thermal runaway in the module shall be the same as that used to initiate and propagate thermal runaway within the module level test		P
9.2.24	The composition, velocity and temperature of the initiating BESS unit vent gases shall be measured within the calorimeter's exhaust duct	Testing conducted outdoor	N/A
9.2.25	The hydrocarbon content of the vent gas shall be measured using flame ionization detection	Testing conducted outdoor	N/A
9.2.26	The test shall be terminated if:		P
	a) Temperatures measured inside each module within the initiating BESS unit return to ambient temperature;	Applicable	P
	b) The fire propagates to adjacent units or to adjacent walls; or		N/A
	c) A condition hazardous to test staff or the test facility requires mitigation		N/A
9.2.27	For residential use systems, the gas collection data gathered in 9.2 shall be compared to the smallest room installation	Non-residential	N/A
9.3	Test method – Outdoor ground mounted units		
9.3.1	Outdoor ground mounted non-residential use BESS being evaluated for installation in close proximity to buildings shall use the test method described in Section 9.2	See 9.2	P
9.3.2	except as noted in 9.3.3 and 9.3.4. Heat flux measurements for the accessible means of egress shall be measured in accordance with 9.2.20.	See 9.2	P
9.3.3	Test samples shall be installed as shown in Figure 9.2 in proximity to an instrumented wall section that is 3.66-m (12-ft) tall with a 0.3-m (1-ft) wide horizontal soffit		P
	Exception: If the manufacturer requires installation against non-flammable material, the test setup may include manufacturer recommended backing material between the unit and plywood wall		N/A
9.3.4	Target BESS shall be installed on each side of the initiating BESS in accordance with the manufacturer's installation specifications	Target unit on the front, side and back of the initiating BESS	P
9.4	Test Method – Indoor wall mounted units	Outdoor	N/A

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
9.4.1	Testing of indoor wall mounted BESS shall be in accordance with Section 9.2, except as modified in this section. See Figure 9.3.		N/A
9.4.2	The test shall be conducted in a standard NFPA 286 fire test room, 3.66 × 2.44 × 2.44-m (12 × 8 × 8-ft) high, with a 0.76 × 2.13-m (2-1/2 × 7-ft) high opening.		N/A
9.4.3	The initiating BESS unit shall be positioned on the wall opposite of the door opening		N/A
9.4.4	Target BESS shall be installed on the wall on each side of the initiating BESS		N/A
9.4.5	The wall on which the initiating and target BESS units are mounted shall be instrumented in accordance with Section 9.2.		N/A
9.4.6	The gas collection methods shall be in accordance with 9.2		N/A
9.4.7	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth ignition indicator.		N/A
9.5	Test Method – Outdoor wall mounted units	Outdoor and ground mounted	N/A
9.5.1	Testing of outdoor wall mounted BESS shall be in accordance with Section 9.2, except as modified in this section. See Figure 9.4.		N/A
9.5.2	Test samples shall be mounted on an instrumented wall section that is 3.66-m (12-ft) tall with a 0.3-m (1-ft) wide horizontal soffit (undersurface of the eave shown in Figure 9.4).		N/A
9.5.3	The initiating BESS unit shall be positioned on the instrumented wall, with its center located 1.22-m (4-ft) above the floor,		N/A
9.5.4	Target BESS shall be installed on the wall on each side of the initiating BESS, at the same height		N/A
9.5.5	The wall on which the initiating and target BESS units are mounted shall be instrumented in accordance with Section 9.2.		N/A
9.5.6	For residential use BESS, the DUT shall be covered with a single layer of cheese cloth		N/A
9.6	Rooftop and open garage installations	Outdoor and ground mounted	N/A
9.6.1	Testing of BESS intended for non-residential use rooftop or open garage installations shall be in accordance with 9.2.		N/A
9.6.2	If intended for rooftop and open garage use only installations, the smoke release rate, the convective and chemical heat release rate and content, velocity and temperature of the released vent gases need not be measured		N/A
9.7	Unit level test report		
9.7.1	The report on the unit level testing shall identify the type of installation being tested, as follows:		P

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
	a) Indoor floor mounted non-residential use BESS; b) Indoor floor mounted residential use BESS; c) Outdoor ground mounted non-residential use BESS; d) Outdoor ground mounted residential use BESS; e) Indoor wall mounted non-residential use BESS; f) Indoor wall mounted residential use BESS; g) Outdoor wall mounted non-residential use BESS; h) Outdoor wall mounted residential use BESS; i) Rooftop installed non-residential use BESS; or j) Open garage installed non-residential use BESS.	Outdoor ground mounted non-residential use BESS;	P
9.7.2	With reference to 9.7.1, if testing is intended to represent more than one installation type, this shall be noted in the report	One installation type only	P
9.7.3	The report shall include the following, as applicable:	See table 1	P

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
	a) Unit manufacturer name and model number (and whether UL 9540 compliant); b) Number of modules in the initiating BESS unit; c) The construction of the initiating BESS unit per 5.3; d) Fire protection features/detection/suppression systems within unit; e) Module voltage(s) corresponding to the tested SOC; f) The thermal runaway initiation method used; g) Location of the initiating module within the BESS unit; h) Diagram and dimensions of the test setup including mounting location of the initiating and target BESS units, and the locations of walls, ceilings, and soffits; i) Observation of any flaming outside the initiating BESS enclosure and the maximum flame extension; j) Chemical and convective heat release rate versus time data; k) Separation distances from the initiating BESS unit to target walls (e. g. distances A and C in Figure 9.1); l) Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and H in Figure 9.1); m) The maximum wall surface and target BESS temperatures achieved during the test and the location of the measuring thermocouple; n) The maximum ceiling or soffit surface temperatures achieved during the indoor or outdoor wall mounted test and the location of the measuring thermocouple; o) The maximum incident heat flux on target wall surfaces and target BESS units; p) The maximum incident heat flux on target ceiling or soffit surfaces achieved during the indoor or outdoor wall mounted test; q) Gas generation and composition data; r) Peak smoke release rate and total smoke release data; at which activation occurred; t) Observation of flying debris or explosive discharge of gases; u) Observation of re-ignition(s) from thermal runaway events; v) Observation(s) of sparks, electrical arcs, or other electrical events; w) Observations of the damage to: 1) The initiating BESS unit; 2) Target BESS units; 3) Adjacent walls, ceilings, or soffits; and x) Photos and video of the test.		
9.8	Performance at unit level testing		

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
9.8.1	Installation level testing in Section 10 is not required if the following performance conditions outlined in Table 9.1 are met during the unit level test.	Conditions in table 9.1 met	P
10	Installation Level	Unit level testing only	N/A
10.1	General		N/A
10.1.1	The installation level test method assesses the effectiveness of the fire and explosion mitigation methods for the BESS in its intended installation		N/A
	a) Test Method 1 – "Effectiveness of sprinklers" is used		N/A
	b) Test Method 2 – "Effectiveness of fire protection plan" is used		N/A
10.1.2	Installation level testing is not appropriate for units only intended for outdoor use or residential use.	Outdoor only	P
10.2	Sample		N/A
10.2.1	The samples (initiating BESS and target BESS) and their preparation for testing		N/A
10.2.2	A flame indicator consisting of a cable tray with fire rated cables that complies with UL 1685 and representative of the installation per the manufacturer's specifications		N/A
10.3	Test method 1 – Effectiveness of sprinklers		N/A
10.3.1	For BESS units with a height of 2.44 m (8 ft) or less, the test shall be conducted in a 6.10 × 6.10 × 3.05-m (20 × 20 × 10-ft) high test room		N/A
10.3.2	The test room shall be fitted with four sprinklers at 3.05-m (10-ft) spacing in the center		N/A
10.3.3	Walls shall be constructed with 16-mm (5/8-in) gypsum wall board		N/A
10.3.4	The initiating BESS unit shall be positioned at manufacturer specified distances		N/A
10.3.5	Temperature measurements at the ceiling locations directly above the initiating and target BESS unit shall be collected by an array of thermocouples		N/A
10.3.6	Instrumented wall surface temperature measurements shall be collected in a vertical array at 152-mm (6-in) intervals		N/A
10.3.7	Thermocouples for wall surface temperature measurements shall be secured to gypsum surfaces by the use of staples		N/A
10.3.8	Heat flux shall be measured with the sensing element of at least two water-cooled Schmidt-Boelter gauges at the surface of each instrumented wall:		N/A
	a) Both are collinear with the vertical thermocouple array;		N/A
	b) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module; and		N/A
	c) One is positioned at the elevation estimated to receive the greatest heat flux during potential propagation of thermal runaway within the initiating BESS unit.		N/A

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
10.3.9	Heat flux shall be measured with at least two sensing water-cooled Schmidt-Boelter gauges at the surface of each adjacent target BESS unit that faces the initiating BESS unit:		N/A
	a) One is positioned at the elevation estimated to receive the greatest heat flux due to the thermal runaway of the initiating module within the initiating BESS; and		N/A
	b) One is positioned at the elevation estimated to receive the greatest surface heat flux due to the thermal runaway of the initiating BESS.		N/A
10.3.10	The heat flux shall be measured with the sensing element of at least one water-cooled Schmidt-Boelter gauge		N/A
10.3.11	No. 24-gauge or smaller Type-K exposed junction thermocouples shall be installed		N/A
10.3.12	An internal fire condition in accordance with the module level test shall be created		N/A
	a) The position of the module shall be selected to present the greatest thermal exposure		N/A
	b) The setup (i.e. type, quantity and positioning) of equipment for initiating thermal runaway in the module shall be the same		N/A
10.3.13	The composition of BESS unit vent gases shall be measured		N/A
10.3.14	The test shall be terminated if:		N/A
	a) Temperatures measured inside each module of the initiating BESS return to below the cell vent temperature;		N/A
	b) The fire propagates to adjacent units or to adjacent walls; or		N/A
	c) A condition hazardous to test staff or the test facility requires mitigation.		N/A
10.3.15	The initiating unit shall be under observation for 24 h after conclusion of the installation test		N/A
10.4	Installation level test report – Test method 1 – Effectiveness of sprinklers	No sprinkler system	N/A
10.4.1	The report on installation level testing shall include the following:		N/A

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
	a) Unit manufacturer name and model number (and whether compliant with UL 9540); b) Number of modules in the initiating BESS unit; c) The construction of the initiating BESS unit per 5.3; d) Module voltage(s) of initiating BESS corresponding to the tested SOC; e) The thermal runaway initiation method used; f) Diagram and dimensions of the test setup including location of the initiating and target BESS units, and the locations of walls and ceilings; g) Location of initiating module within the BESS unit; h) Separation distances from the initiating BESS unit to (e.g. distances A and C in Figure 10.1); i) Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and E in Figure 10.1); j) Distances of the flame indicator (if used) with respect to the BESS (e. g. distances A and B in Figure 10.2); k) Maximum temperature at the ceiling; l) Distance of fire spread within the flame indicator; m) The maximum wall surface and target BESS unit temperatures achieved during the test and the location of the measuring thermocouple; n) The maximum incident heat flux on target wall surfaces and target BESS units; o) Voltages of initiating BESS; p) Total number of sprinklers that operated and length of time the sprinklers operated during the test; q) Gas generation and composition data, if measured; r) Observation of flaming outside of the test room s) Observation of flying debris or explosive discharge of gases; t) Observation of re-ignition(s) from thermal runaway events; u) Observations of the damage to: 1) The initiating BESS unit; 2) Target BESS units; and 3) Adjacent walls; v) Photos and video of the test; w) Fire protection features/detection/suppression systems within unit; and x) Sprinkler K-factor, RTI, manufacturer and model, number of sprinklers and layout		N/A
10.5	Performance – Test method 1 – Effectiveness of sprinklers		N/A
10.5.1	For BESS units intended for installation in locations with combustible construction, surface temperature measurements along instrumented wall surfaces shall not exceed a temperature rise of 97°C		N/A
10.5.2	The surface temperature of modules within the BESS units adjacent to the initiating BESS unit shall not exceed the temperature at which thermally initiated cell venting occurs		N/A

Report No. NN22GLCI.001

ANSI/CAN/UL 9540A:2019			
Clause	Requirement – Test	Result – Remark	Verdict
10.5.3	The fire spread on the cables in the flame indicator shall not extend horizontally beyond the initiating BESS enclosure dimensions		N/A
10.5.4	There shall be no flaming outside the test room.		N/A
10.5.5	There is no observation of detonation.		N/A
10.5.6	Heat flux in the center of the accessible means of egress shall not exceed 1.3 kW/m ² .		N/A
10.5.7	There shall be no observation of re-ignition within the initiating unit after the installation test		N/A
10.5.8	An installation level test that does not meet the applicable performance criteria noted above is considered noncompliant and would need to be revised and retested		N/A
10.6	Test method 2 – Effectiveness of fire protection plan	Refer 10.3	N/A
10.6.1	The test method 2 test set-up and test procedures are identical to that in 10.3, except instead of the sprinkler system set up of 10.3.2, the room shall be fitted with the specified fire protection		N/A
10.7	Installation level test report – Test method 2 – Effectiveness of fire protection plan	Refer 10.4	N/A
10.7.1	The report on installation level testing shall include the following:		N/A
	a) The report information in 10.4.1 items (a) – (u), and (v) if applicable; b) Fire protection features/detection/suppression systems within installation; and c) Length of time of operation of the clean agent, or other suppression system in addition to any sprinklers used.		N/A
10.8	Performance – Test method 2 – Effectiveness of fire protection plan	Refer 10.5	N/A
10.8.1	See 10.5 for performance criteria		N/A

ANNEX A	Test Concepts And Application Of Test Results To Installations	INFORMATIVE	
A2.1	General		N/A
A2.2	Cell level testing	CATL UL9540a certified by UL project no. 4789956341	P
A2.3	Module level testing	CATL UL9540a certified by TÜV Sud project no. 64.290.21.30597.01	P
A2.4	Unit level testing	Refer Table 1 below for results	P

Report No. NN22GLCI.001

ANNEX B	Safety Recommendations for Testing	INFORMATIVE	
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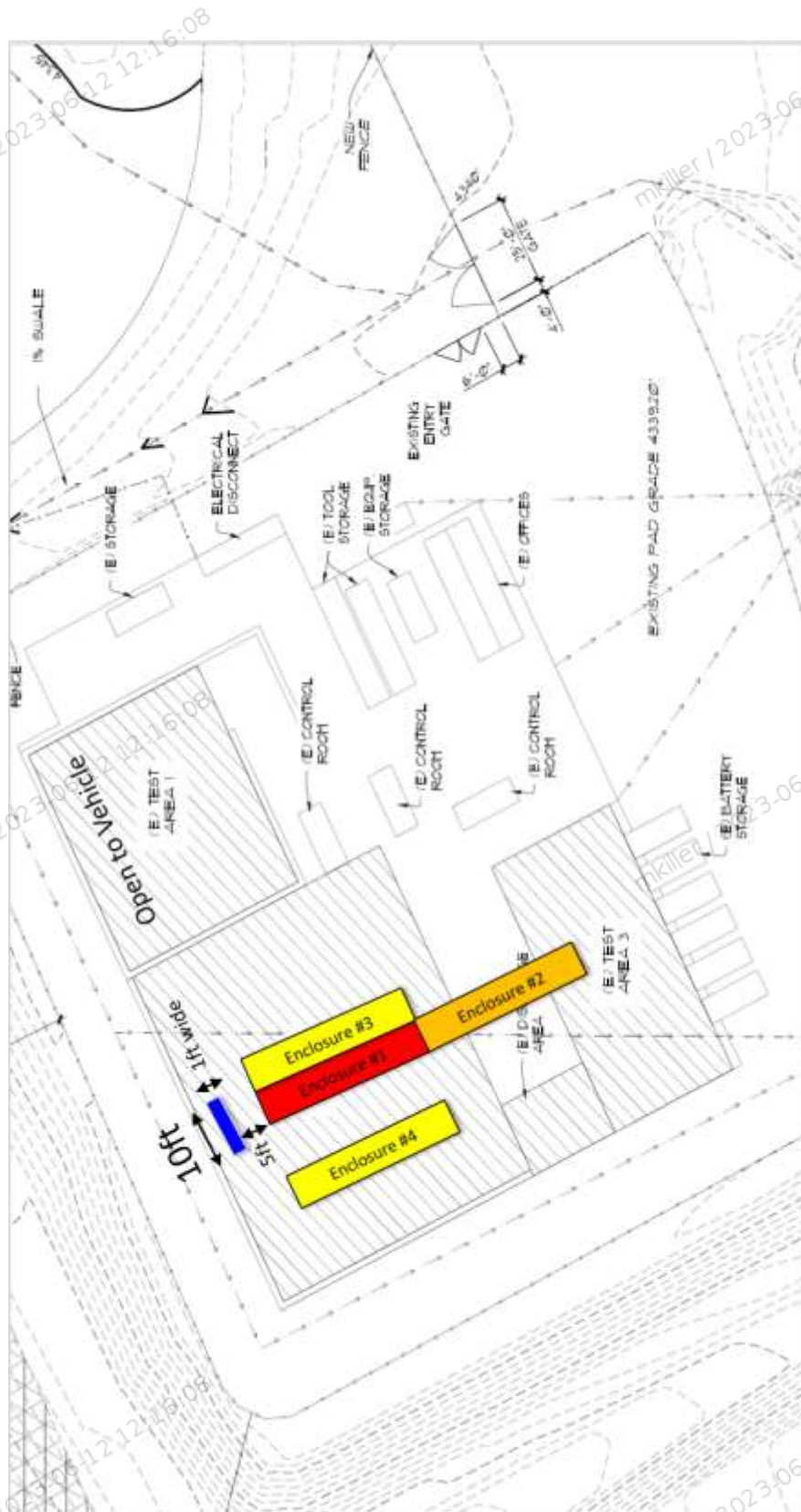


Fig 1. Site Layout

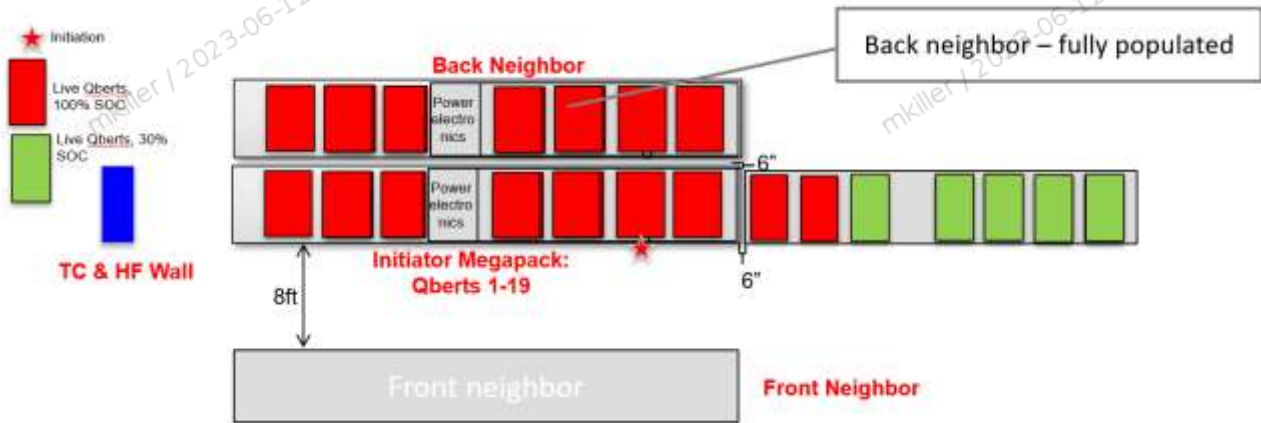


Fig 2. Test area Layout

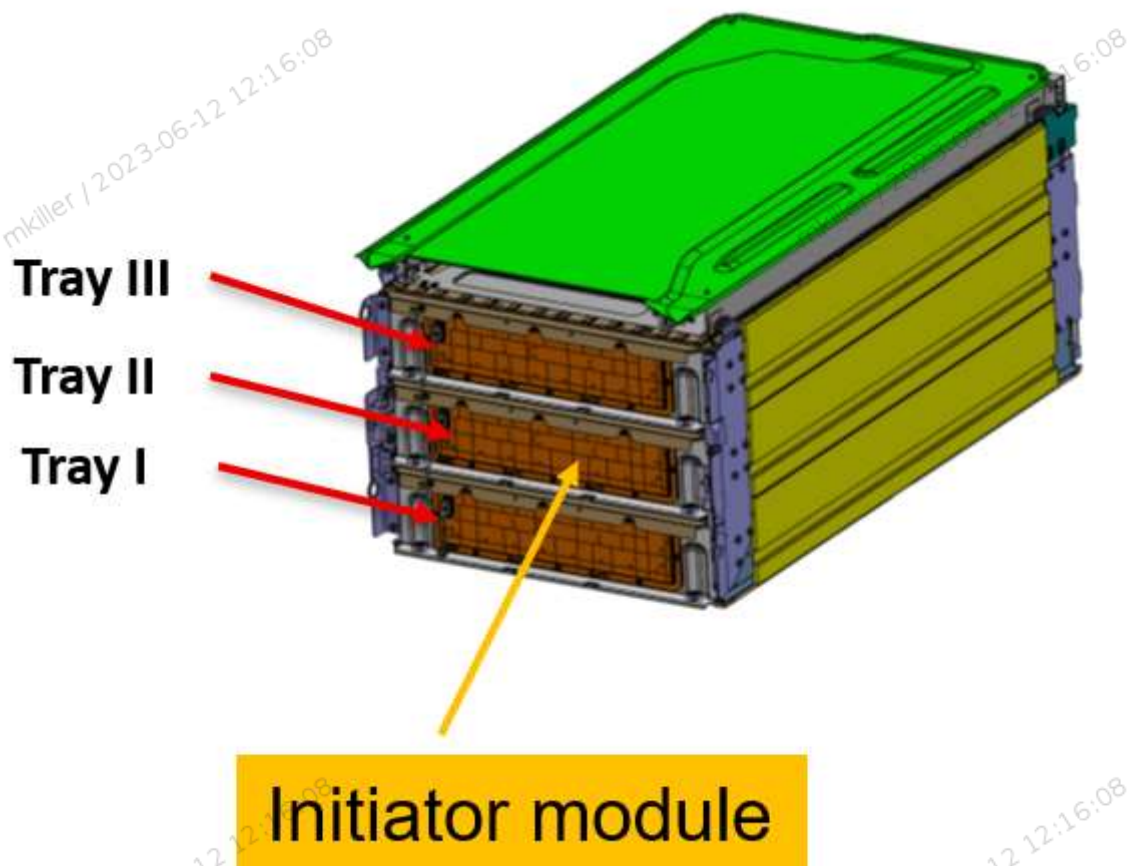


Fig 3. Initiator module location



Fig 4. Initiator AC Battery module location

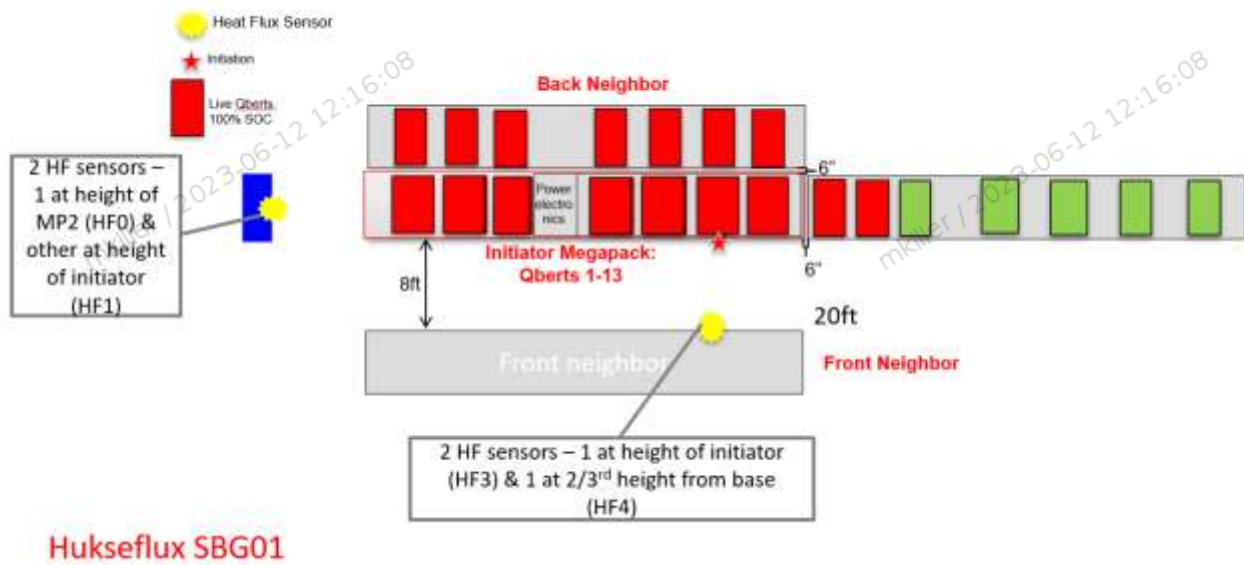


Fig 5. Heat Flux locations



Report No. NN22GLCI.001



Fig 6. Camera Layout



Fig 7. Pre test photos



Fig 8. Post-test photo of Initiator location

Report No. NN22GLCI.001



Fig 9. Internal view, post test photo



Fig 10. Initiator Qbert module, Rear view – Post test



Fig 11. Front view – Post test

Report No. NN22GLCI.001



Fig 12. Side view

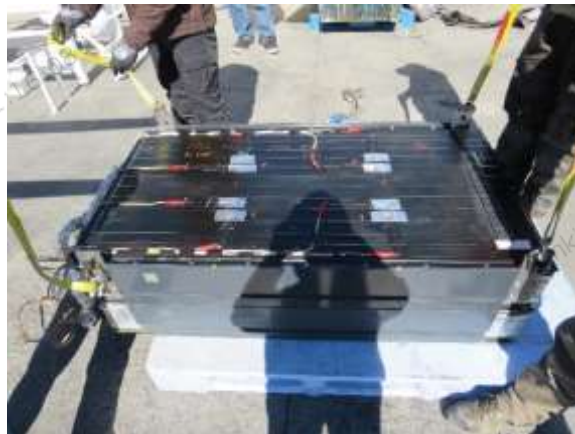


Fig 13.– Initiator Battery module – Post test

Report No. NN22GLCI.001

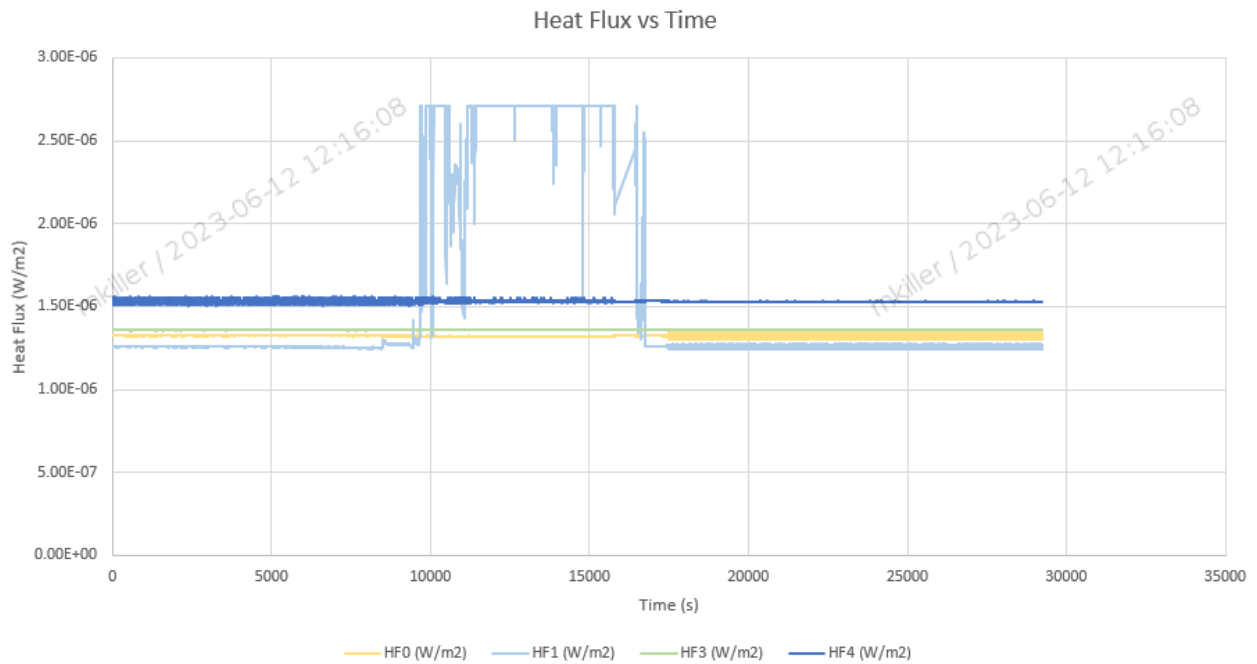


Fig 14.- Heat Flux graph

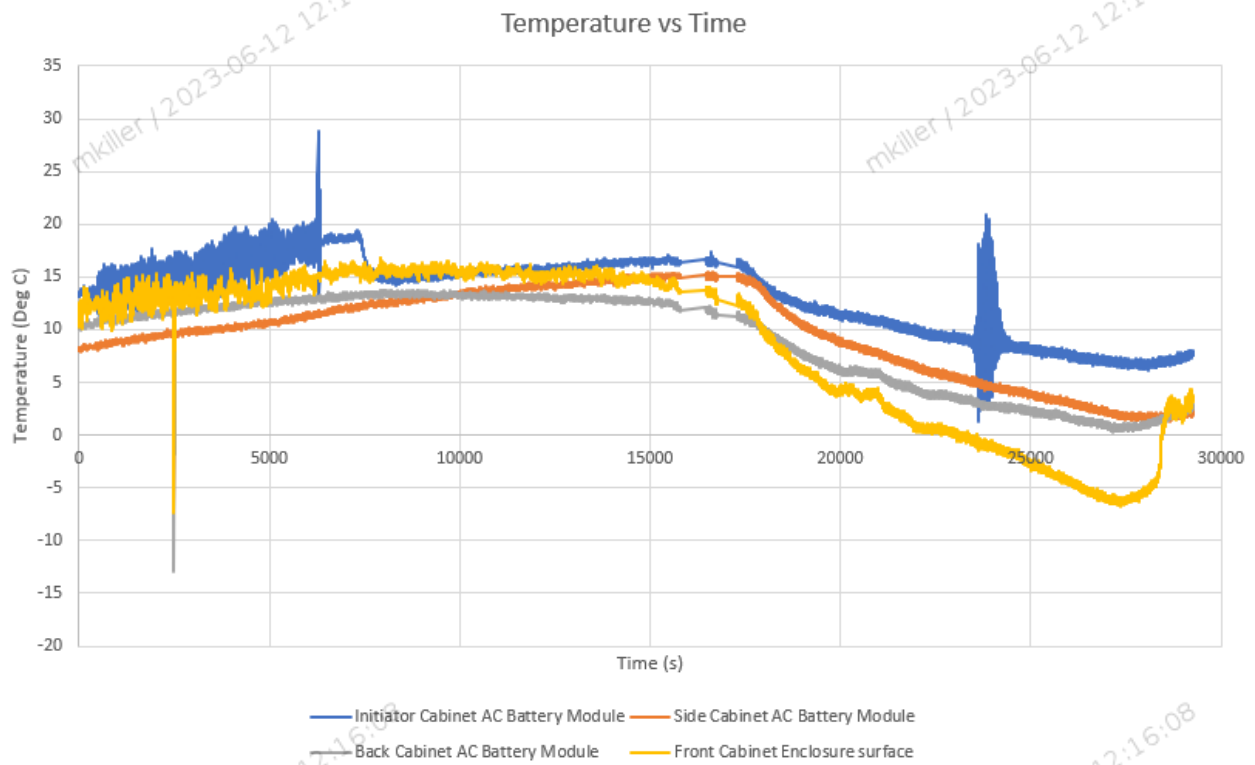


Fig 15.- Temperature graph – Target BESS and Initiator BESS non-initiating AC Battery modules

Report No. NN22GLCI.001

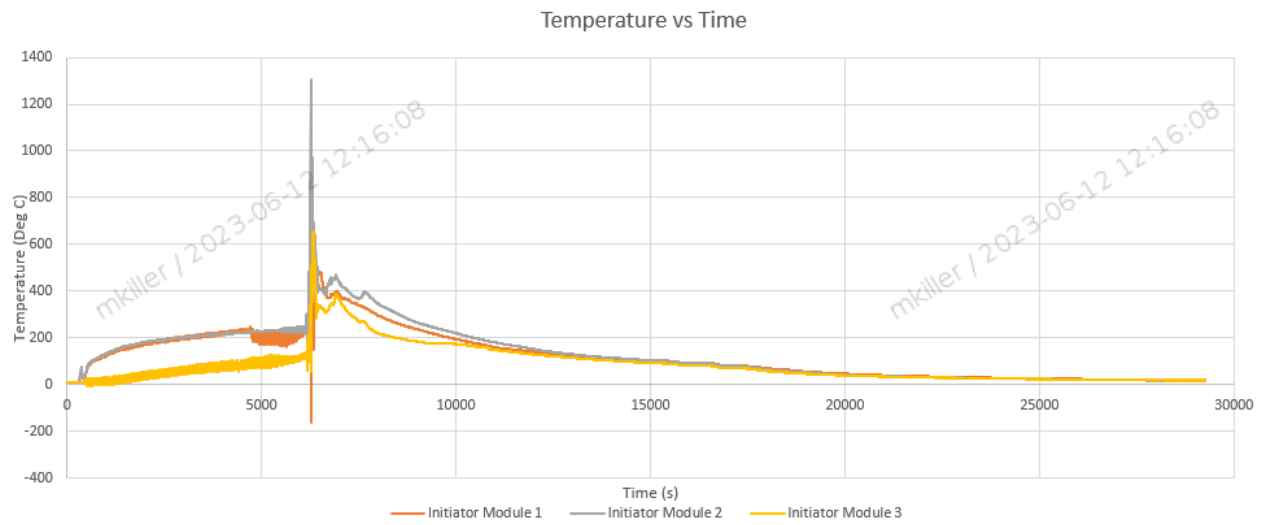


Fig 16.- Temperature graph – Initiator module tray

Table 1. Test results per Clause 9.7

#	Items	Description	Comment
a)	Unit manufacturer name and model number (and whether UL 9540 compliant);	Tesla, Megapack 2, 174884-XX-Y, UL9540 compliant	Pass
b)	Number of modules in the initiating BESS unit;	3 module forming AC Battery and 19 AC Batteries in Megapack	Pass
c)	The critical construction of the initiating BESS unit per 5.3;	UL 9540 compliant.	Pass
d)	Fire protection features/detection/suppression systems within unit;	Optional Detection Signal	Pass
e)	Module voltage(s) corresponding to the tested SOC;	100%SOC measured. 373.5V Module voltage	Pass
f)	The thermal runaway initiation method used;	4 film heaters heating 6 cells simultaneously	Pass
g)	Location of the initiating module within the BESS unit;	Initiator Megapack AC Battery 7-3	Pass
h)	Diagram and dimensions of the test setup including mounting location of the initiating and target BESS units, and the locations of walls, ceilings, and soffits;	6 inches from ISO knuckle of Initiating unit to side and back Target unit. 8 ft from front Target.; wall – 5 ft from Initiator	Pass
i)	Observation of any flaming outside the initiating BESS enclosure;	No	Pass
j)	Chemical and convective heat release rate versus time data;	N/A	N/A
k)	Separation distances from the initiating BESS unit to target walls (e.g. distances A and C in Figure 9.1);	5 ft	Pass
l)	Separation distances from the initiating BESS unit to target BESS units (e.g. distances D and H in Figure 9.1);	6 inches from ISO knuckle	Pass
m)	The maximum wall surface and target BESS temperatures achieved during the test and the location of the measuring thermocouple;	Wall – 30.75 deg C Front Target surface- 16.8C	Pass
n)	The maximum ceiling or soffit surface temperatures achieved during the indoor or outdoor wall mounted test and the location of the measuring thermocouple;	N/A	N/A
o)	The maximum incident heat flux on target wall surfaces and target BESS units;	Wall- 2.71×10^{-6} W/m ² Front Target – 1.54×10^{-6} W/m ²	Pass
p)	The maximum incident heat flux on target ceiling or soffit surfaces achieved during the indoor or outdoor wall mounted test;	N/A	N/A
q)	Gas generation and composition data;	N/A	N/A
r)	Peak smoke release rate and total smoke release data;	N/A	N/A
s)	Indication of the activation of integral fire protection systems and if activated the time into the test at which activation occurred;	N/A	N/A
t)	Observation of flying debris or explosive discharge of gases;	None observed	Pass
u)	Observation of re-ignition(s) from thermal runaway events;	None observed	Pass
v)	Observation(s) of sparks, electrical arcs, or other electrical events;	None observed	Pass
w)	Observations of the damage to: 1) The initiating BESS unit;	1. No external damage. Initiating AC battery	Pass

Report No. NN22GLCI.001

	2) Target BESS units; and 3) Adjacent walls	module had burn marks. Other 2 modules of AC Battery intact. Balance 18 AC AC Battery intact. 2. None observed 3. None observed	
x)	Photos and video of the test	Attached	Pass
<u>Note:</u>			

- End of Report -